

ECON 453: Data Analytics and Modeling: Quantitative Analysis for Economic Strategy

Fall 2026

The University of Arizona

Department of Economics

Instructor: Tu Li

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Instructor Office Hours: Thursday, 3:30–4:30 p.m.

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Class Meetings: Tuesdays and Thursdays, 2:00–3:15 p.m.

Class Location: McClelland Hall 133

Course Description

This course introduces core statistical tools and modern machine learning methods used to analyze business and economic data. Topics include linear regression, logistic regression, cross-validation, bootstrap methods, nonparametric regression, high-dimensional regression, and random forests. Students will learn to apply these methods using computer software, evaluate model performance, interpret results, and communicate findings in practical business settings.

Foundational concepts in probability and statistics will be introduced as needed, including probability distributions, descriptive statistics, data cleaning and visualization, confidence intervals, hypothesis testing, and maximum likelihood estimation. The course emphasizes intuition and application, with particular attention to the connection between traditional statistical inference and modern predictive methods.

Expected Learning Outcomes

Upon successful completion of this course, students will be able to:

- clean, visualize, and summarize business and economic data;
- select and apply appropriate statistical and machine learning methods for prediction and decision-making;
- conduct basic statistical inference and assess estimation uncertainty;
- implement course methods using computer software; and
- interpret and communicate results from statistical and machine learning models.

Prerequisites

Prerequisites are (ECON 300 or ECON 361) and (AREC/ECON 339 or BNAD 276). Students should be familiar with basic probability, statistics, and algebra. Any additional matrix-algebra concepts needed for the course will be introduced as they arise. No prior programming experience is required.

Required Textbook

The required textbook is Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, *An Introduction to Statistical Learning: with Applications in R*, 2nd ed. (Springer, 2021), hereafter **ISLR**. The lecture slides are designed to be self-contained, but students are encouraged to read the assigned sections before class. The book is freely available from the authors' website.

Computer Software

Computer software will be used for instruction, demonstrations, and selected course activities. No prior programming experience is required.

How to Succeed in This Course

This course emphasizes intuition, application, and implementation rather than mathematical theory. Even so, learning data analysis requires sustained practice outside the classroom. Depending on your background, you should expect to spend approximately **5–15 hours** per week reviewing examples, completing coursework, and practicing with computer software.

Grading Policy

The final course grade will be based on the following components:

- **Homework 1 (5%):** Probability.
- **Homework 2 (5%):** Statistical Inference.
- **Homework 3 (5%):** Simple Linear Regression.
- **Homework 4 (5%):** Matrix Algebra and Multiple Regression.
- **Homework 5 (5%):** Regularization, Flexible Models, and Classification.
- **Exam 1 (25%):** In class.
- **Exam 2 (25%):** In class.
- **Exam 3 (25%):** Cumulative; Monday, December 14, 3:30–5:30 p.m.

Homework deadlines. Homework 1, 2, and 3 will be due before Exam 1, and Homework 4 and 5 will be due before Exam 2. Exact due dates will be announced separately.

Exam timing. The dates for Exams 1 and 2 shown in the tentative schedule are provisional and may change based on course progress. Exam 1 is planned after Lecture 7, and Exam 2 is planned after Lecture 12. Exam 3 is cumulative and covers material from the entire course.

Grading scale: The minimum percentages for each letter grade are A: 90%, B: 80%, C: 70%, and D: 60%. Scores below 60% receive an E.

Late work will not be accepted unless an approved accommodation applies.

Tentative Schedule

ISLR refers to the required R textbook listed above. Reading numbers refer to sections of ISLR. A blank entry means that no ISLR reading is assigned.

Week	Dates	Topic	Reading
1	Aug. 25 and 27	Probability I	
2	Sept. 1 and 3	Probability II	
3	Sept. 8 and 10	Statistics I	
4	Sept. 15 and 17	Statistics II	
5	Sept. 22 and 24	Overview of Statistical Learning	ISLR 2.1
6	Sept. 29 and Oct. 1	Simple Linear Regression (OLS)	ISLR 3.1.1
7	Oct. 6 and 8	Model Inference	ISLR 3.1.2
8	Oct. 13 and 15	Review and Exam 1 (tentative)	
9	Oct. 20 and 22	Matrix Algebra	
10	Oct. 27 and 29	Multiple Regression	ISLR 3.2, 3.3.1–3.3.2
11	Nov. 3 and 5	Regularization	ISLR 2.2.2, 6.1–6.2, 6.4
12	Nov. 10 and 12	Nonparametric Regression	ISLR 7.1, 7.3
13	Nov. 17 and 19	Review and Exam 2 (tentative)	
14	Nov. 24; Nov. 26 (no class)	Catch up	
15	Dec. 1 and 3	Logistic Regression	ISLR 4.2–4.3
16	Dec. 8	Resampling Methods	ISLR 5.1–5.2
Exam 3	Mon., Dec. 14, 3:30–5:30 p.m.	Cumulative Exam	

Thursday, December 10, is Reading Day; no classes or final exams will be held.

University Policies

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss the principles and applications of course materials. Submitted work must reflect each student's own understanding. Unless collaboration is explicitly authorized, students must prepare their own submissions. Any permitted use of AI tools must follow the instructions for the assignment, and students remain responsible for checking their work. Students are expected to follow the UA Code of Academic Integrity.

Accommodations for Students with Disabilities

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center at 520-621-3268 to establish reasonable accommodations.

Honors Contract

Students wishing to contract this course for honors credit should email the instructor to discuss the terms of the contract. Information is available on the Honors Contracts website.

Safety on Campus and in the Classroom

For emergency procedures covering a range of incidents, please visit the Critical Incident Response Team (CIRT) website.

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself.

Nondiscrimination and Anti-harassment Policy

The University of Arizona is committed to creating and maintaining an environment free of discrimination. The University prohibits discrimination, harassment, and retaliation based on protected classifications, including race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information. See the Nondiscrimination and Anti-Harassment Policy.

Absence and Class Participation Policies

Reasonable accommodations will be provided for absences resulting from sincerely held religious beliefs, observances, or practices. Refer to the Religious Accommodation Policy. Absences pre-approved by the University Dean of Students (or dean's designee) will be honored.

Additional University Policies

All university policies related to a syllabus are available on the UA syllabus policies page.

Additional Resources for Students

See the University of Arizona academic policies and procedures and Student Assistance and Advocacy.

Subject to Change Notice

Information in this syllabus, other than the grade and absence policies, may be changed with reasonable advance notice when deemed appropriate by the instructor.